THE MULTIPLICATIVE AUTOCORRELATION OF THE FRACTIONAL PART HAS NO LOCAL MAXIMUM AT IRRATIONAL POINTS

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ABSTRACT. Let $A(\lambda) = \int_0^\infty \{t\}\{\lambda t\} t^{-2}\,\mathrm{d}t$ be the multiplicative autocorrelation function of the fractional part, which gives the Gram matrix of the Nyman–Beurling system $e_\alpha(t) = \{t/\alpha\}$. Báez-Duarte, Balazard, Landreau and Saias proved that A has a strict local maximum at every positive rational, and asked whether A has a strict local maximum at some positive irrational. We show that it does not: at every positive irrational λ the symmetric second difference $A(\lambda+h)+A(\lambda-h)-2A(\lambda)$ is positive along a sequence $h \to 0^+$ determined by the continued fraction of λ, so A has no local maximum at λ, strict or not. Hence the local maxima of A on $(0,\infty)$ are exactly the positive rationals. The proof combines the exact functional equation of Báez-Duarte, Balazard, Landreau and Saias for $\varphi_2(x) = \sum_{k\ge1} B_2(kx)/k^2$ near a rational with the observation that its remainder term is unchanged when φ_2 is shifted by a constant.

1. INTRODUCTION

Let $\{x\}$ denote the fractional part of the real number x, and let L denote the Hilbert space $L^2(0,\infty; t^{-2}\,\mathrm{d}t)$, equipped with

$$\|f\|^2 = \int_0^\infty |f(t)|^2 \frac{\mathrm{d}t}{t^2}.$$

Put $\chi(t) = [t \ge 1]$ and $e_\alpha(t) = \{t/\alpha\}$ for $\alpha \ge 1$. Nyman's criterion [9] (see also Beurling [7]), in the form stated in [3, Section 1], asserts that the Riemann hypothesis holds if and only if χ lies in the closed linear span in L of the functions e_α, $\alpha \ge 1$, and Báez-Duarte [2] showed that it suffices to use integer α. Trying to approximate χ by linear combinations of the e_α leads to the inner products

$$\langle e_\alpha, e_\beta\rangle = \int_0^\infty \{t/\alpha\}\{t/\beta\}\frac{\mathrm{d}t}{t^2} = \frac{1}{\alpha} A(\alpha/\beta),$$

where

$$A(\lambda) = \int_0^\infty \{t\}\{\lambda t\}\frac{\mathrm{d}t}{t^2}, \qquad \lambda \ge 0,$$

Date: October 13, 2026.

2020 *Mathematics Subject Classification.* Primary 11J70; Secondary 26A27, 11M26.

Key words and phrases. Fractional part, multiplicative autocorrelation, Nyman–Beurling criterion, continued fractions, approximate functional equations.

is the *multiplicative autocorrelation function* of the fractional part (see [3, Section 2]). The identity follows from the substitution $t \mapsto \alpha t$.

The function A has a rich arithmetic structure. Vasyunin [10] found an explicit formula for $A(h/k)$ in terms of the cotangent sum

$$V(h,k) = \sum_{j=1}^{k-1} \{jh/k\} \cot(j\pi/k).$$

Báez-Duarte, Balazard, Landreau and Saias [1] showed that A is continuous on $[0,\infty)$, that it is not differentiable at any positive rational, and that it has a strict local maximum at every positive rational. Balazard and Martin [4, Théorème 1] later proved that A is differentiable at λ if and only if λ is an irrational number such that the series $\sum_{k\geq 0}(-1)^k(\log q_{k+1})/q_k$ converges, where q_k are the denominators of the convergents of λ; see also de la Bretèche and Tenenbaum [6]. In his survey at the Oberwolfach workshop *Dirichlet Series and Function Theory in Polydiscs* (2014), Balazard recalled the following question from [1].

Problem 1 (Balazard [3, Question 2])**.** *Does there exist a positive irrational where A has a strict local maximum?*

The main purpose of the present paper is to answer this question in the negative. For an irrational number λ with convergents p_k/q_k we put

$$h_k = |\lambda - p_k/q_k|.$$

Theorem 1. *Suppose that $\lambda > 0$ is irrational. Then there are infinitely many k such that*

$$A(\lambda + h_k) + A(\lambda - h_k) - 2A(\lambda) > 0. \tag{1}$$

In particular, A does not have a local maximum at λ, strict or not.

The second statement follows from the first, since $h_k \to 0$ and (1) gives $\max\{A(\lambda+h_k), A(\lambda-h_k)\} > A(\lambda)$. Combining Theorem 1 with the result of Báez-Duarte, Balazard, Landreau and Saias at rational points, we obtain a complete description of the local maxima of A.

Corollary 1. *The set of points in $(0,\infty)$ at which A has a local maximum is $\mathbb{Q} \cap (0,\infty)$, and each of these local maxima is strict.*

To the best of our knowledge, Theorem 1 is new. The proof gives a quantitative form of (1), which we state in Section 5.

Let us briefly describe our approach. The function A is, up to a smooth error, equal to $\varphi_2(\lambda)/(2\lambda)$, where $\varphi_2(x) = \sum_{k\geq 1} B_2(kx)/k^2$ and B_2 is the second Bernoulli function [1, Proposition 90]. Near a rational p/q, the function φ_2 has the expansion [1, Proposition 78]

$$\varphi_2(p/q + t) - \varphi_2(p/q) = \frac{|t|\log|t|}{q} + c_{\pm}t - \frac{t^2}{2} + O((qt)^3), \tag{2}$$

where $c_\pm$ depends on p, q and the sign of t. The term $|t|\log|t|/q$ produces a sharp peak at p/q; this is the source of the strict local maxima at rationals. However, the same term is *convex* on each side of p/q, since $(t\log t)'' = 1/t > 0$. An irrational λ at distance h from a convergent p/q sits on one flank of this peak, and the second difference of $|t|\log|t|/q$ at the points $0, h, 2h$ equals $(2\log 2)h/q > 0$. To make this rigorous, we must show that the remainder in (2) is smaller than $(2\log 2)h/q$. The remainder is small relative to the main term only when q^2h is small, that is, when the partial quotient following p/q is large. With (2) alone, this argument therefore covers only irrationals with unbounded partial quotients, and not, for example, the golden ratio. To handle every irrational we use the exact form of (2), namely [1, Proposition 77], in which the remainder is expressed through φ_2 itself near the point $\bar{p}/q$, where $p\bar{p} \equiv 1 \pmod q$. The new observation is that this remainder does not change when φ_2 is shifted by a constant. Centring φ_2 halves the explicit bound for the remainder, and the resulting inequality holds as soon as $q^2h < 1/2$. Every irrational has infinitely many convergents with this property.

The present paper is organized into four additional sections. In Section 2 we collect the facts about φ_2 and A from [1] that we use. In Section 3 we bound the second difference of φ_2 from below near a rational number. In Section 4 we transfer this bound to A and prove Theorem 1 and Corollary 1. Finally, Section 5 contains some concluding remarks.

2. Preliminaries

We use the notation of [1], whose results we cite with the numbering of the detailed working document arXiv:math/0306251. Let $B_2(x) = \{x\}^2 - \{x\} + 1/6$ be the second Bernoulli function, and put

$$\varphi_2(x) = \sum_{k\ge 1} \frac{B_2(kx)}{k^2}, \qquad \Delta_{p,q}(t) = \varphi_2(p/q + t) - \varphi_2(p/q).$$

The function φ_2 is continuous and 1-periodic [1, Proposition 67]. Since $-1/12 \le B_2(x) \le 1/6$ for every real x and $\sum_{k\ge1} k^{-2} = \pi^2/6$, we have

$$-\frac{\pi^2}{72} \le \varphi_2(x) \le \frac{\pi^2}{36}, \qquad x \in \mathbb{R}. \tag{3}$$

The midpoint of this interval is $\pi^2/144$ and its half-length is $\pi^2/48$. Let

$$\omega(\delta;\varphi_2) = \sup_{|x-y|\le\delta} |\varphi_2(x) - \varphi_2(y)|$$

be the modulus of continuity of φ_2. By [1, Proposition 74],

$$\omega(\delta;\varphi_2) \ll \delta\log(1/\delta), \qquad 0 < \delta \le 1/2. \tag{4}$$

The link between A and φ_2 is the following identity, valid for every $\lambda > 0$ [1, Proposition 90]:

$$A(\lambda) = \frac{1}{2}\log\lambda + \frac{1-\gamma+\log 2\pi}{2} + \frac{\varphi_2(\lambda)}{2\lambda} - \lambda\int_\lambda^\infty \varphi_2(t)\frac{\mathrm{d}t}{t^3}. \tag{5}$$

Here γ is Euler's constant. All terms except $\varphi_2(\lambda)/(2\lambda)$ are differentiable, and we will see in Lemma 3 that their contribution to second differences is negligible.

The main tool is the exact functional equation of φ_2 near a rational number. Let p and q be coprime integers with $q \geq 1$, and let $\bar{p}$ be an integer with $p\bar{p} \equiv 1 \pmod q$. By [1, Proposition 77], for $t > 0$ and $\varepsilon = \pm 1$ we have

$$\Delta_{p,q}(\varepsilon t) = \frac{t\log t}{q} + \beta_\varepsilon t - \frac{t^2}{2} + R_\varepsilon(t), \tag{6}$$

where $\beta_\varepsilon = \big(\varepsilon\pi V(p,q) + 2\log q + \log 2\pi - \gamma - 1\big)/q$ does not depend on t, and

$$R_\varepsilon(t) = (qt)^3\Delta_{\bar{p},q}(\varepsilon/q^2 t) - 2q^3\int_0^t u(3u-t)\Delta_{\bar{p},q}\big(\varepsilon/q^2 u\big)\,\mathrm{d}u. \tag{7}$$

The functional equation (6) is a consequence of the functional equation of an Estermann-type function attached to p/q, which relates the values at p/q and at $\bar{p}/q$ [1, Propositions 64 and 76]. It should be read as a renormalization: the behaviour of φ_2 at distance t from p/q is described, up to the explicit terms, by φ_2 at distance $1/(q^2 t)$ from $\bar{p}/q$. The asymptotic formula (2) follows from (6) by bounding the remainder crudely [1, Proposition 78]. We will use only that $\Delta_{\bar{p},q}$ is a bounded function and the precise form of (7), not the arithmetic of $\bar{p}$.

Finally, we recall some facts about continued fractions. Suppose that the irrational number λ has the continued fraction expansion $\lambda = [a_0; a_1, a_2, \ldots]$, and let p_k/q_k be its convergents. We put $p_{-1} = 1$ and $q_{-1} = 0$; recall that $p_0 = a_0$ and $q_0 = 1$. Then

$$p_k = a_k p_{k-1} + p_{k-2}, \qquad q_k = a_k q_{k-1} + q_{k-2}, \qquad p_k q_{k-1} - p_{k-1}q_k = (-1)^{k-1} \tag{8}$$

for $k \geq 1$ [8, Theorems 149 and 150]; with our choice of p_{-1} and q_{-1}, the last identity also holds for $k = 0$. In particular, $q_{k-1} \leq q_k$ for $k \geq 1$, and $q_k \to \infty$. We let $x_{k+1} = [a_{k+1}; a_{k+2}, \ldots]$ denote the complete quotients, so that $\lambda = [a_0; a_1, \ldots, a_k, x_{k+1}]$.

3. The second difference of φ_2 near a rational

In this section p and q are coprime integers with $q \geq 1$, and ε, $\bar{p}$ and R_ε are as in (6) and (7). We first record the observation on which the proof rests.

Lemma 1. *Suppose that $t > 0$. Then $R_\varepsilon(t)$ is unchanged if $\Delta_{\bar p,q}$ is replaced by $\Delta_{\bar p,q} + c$ in (7), for any constant c. Consequently,*

$$|R_\varepsilon(t)| \le \frac{7\pi^2}{162} q^3 t^3.$$

Proof. Replacing $\Delta_{\bar p,q}$ by $\Delta_{\bar p,q} + c$ in (7) changes $R_\varepsilon(t)$ by

$$c(qt)^3 - 2cq^3 \int_0^t u(3u - t)\,\mathrm{d}u = cq^3t^3 - 2cq^3 \cdot \frac{t^3}{2} = 0,$$

since $\int_0^t (3u^2 - tu)\,\mathrm{d}u = t^3 - t^3/2 = t^3/2$. Choose $c = \varphi_2(\bar p/q) - \pi^2/144$. Then

$$G(s) \coloneqq \Delta_{\bar p,q}(s) + c = \varphi_2(\bar p/q + s) - \frac{\pi^2}{144},$$

and (3) shows that $|G(s)| \le \pi^2/48$ for every real s. Hence

$$|R_\varepsilon(t)| \le \frac{\pi^2}{48}\Big(q^3t^3 + 2q^3 \int_0^t u|3u - t|\,\mathrm{d}u\Big).$$

Splitting the integral at $u = t/3$, we find that

$$\int_0^t u|3u - t|\,\mathrm{d}u = \int_0^{t/3} u(t - 3u)\,\mathrm{d}u + \int_{t/3}^t u(3u - t)\,\mathrm{d}u = \frac{t^3}{54} + \frac{28t^3}{54} = \frac{29t^3}{54}.$$

Therefore $|R_\varepsilon(t)| \le (\pi^2/48)(1 + 29/27)q^3t^3 = (7\pi^2/162)q^3t^3$. □

Without the shift, the same argument would use only $|\Delta_{\bar p,q}| \le \pi^2/24$, which is the full length of the interval in (3), and would give twice the constant. The factor 2 is what allows us to treat all irrationals in Section 4.

We now bound the second difference of φ_2 at a point λ near p/q, with step equal to the distance from λ to p/q. The relevant parameter is $y = 1/(q^2|\lambda - p/q|)$, which is large when λ is very close to p/q compared with $1/q^2$.

Lemma 2. *Suppose that $d \ne 0$ is real, and put $\lambda = p/q + d$, $\delta = |d|$ and $y = 1/(q^2\delta)$. Then*

$$\varphi_2(\lambda + \delta) + \varphi_2(\lambda - \delta) - 2\varphi_2(\lambda) \ge \frac{\delta}{q}\Big(2\log 2 - \frac{35\pi^2}{81y^2} - \frac{1}{qy}\Big).$$

Proof. Let ε be the sign of d. The three points $\lambda - \delta$, λ and $\lambda + \delta$ are, in some order, the points $p/q + \varepsilon t$ with $t = 0$, $t = \delta$ and $t = 2\delta$; the point λ corresponds to $t = \delta$. Thus

$$\varphi_2(\lambda + \delta) + \varphi_2(\lambda - \delta) - 2\varphi_2(\lambda) = \Delta_{p,q}(2\varepsilon\delta) + \Delta_{p,q}(0) - 2\Delta_{p,q}(\varepsilon\delta).$$

Note that $\Delta_{p,q}(0) = 0$. We insert (6) with $t = 2\delta$ and $t = \delta$. The linear terms cancel, since $\beta_\varepsilon \cdot 2\delta - 2\beta_\varepsilon\delta = 0$. The logarithmic terms give

$$\frac{2\delta\log 2\delta - 2\delta\log\delta}{q} = \frac{2\delta\log 2}{q},$$

and the quadratic terms give $-(4\delta^2 - 2\delta^2)/2 = -\delta^2$. Hence

$$\varphi_2(\lambda+\delta) + \varphi_2(\lambda-\delta) - 2\varphi_2(\lambda) = \frac{2\delta\log 2}{q} - \delta^2 + R_\varepsilon(2\delta) - 2R_\varepsilon(\delta).$$

By Lemma 1,

$$|R_\varepsilon(2\delta)| + 2|R_\varepsilon(\delta)| \le \frac{7\pi^2}{162}(8+2)q^3\delta^3 = \frac{35\pi^2}{81}q^3\delta^3.$$

Finally, $q^3\delta^3 = (\delta/q)(q^2\delta)^2 = (\delta/q)y^{-2}$ and $\delta^2 = (\delta/q)q\delta = (\delta/q)(qy)^{-1}$. □

In Lemma 2, the term $(2\log 2)\delta/q$ is the convexity of the flank of the peak at p/q, and the term $35\pi^2/(81y^2)$ measures how much the peaks at the other rationals near λ can interfere at the scale δ. Since $35\pi^2/324 = 1.0661\ldots$ and $2\log 2 = 1.3862\ldots$, the bound in Lemma 2 is positive as soon as $y > 2$ and q is large.

4. Proof of Theorem 1

We first show that the smooth terms in (5) do not affect second differences at the scales we need. For a function f we write

$$\Delta^2 f(\lambda; h) = f(\lambda+h) + f(\lambda-h) - 2f(\lambda).$$

Lemma 3. *Suppose that $\lambda > 0$. Then there is a constant C_λ such that*

$$\left|\Delta^2 A(\lambda;h) - \frac{\Delta^2\varphi_2(\lambda;h)}{2\lambda}\right| \le C_\lambda h^2\log(1/h)$$

for $0 < h \le \min(\lambda/2, 1/4)$.

Proof. By (5) we have $A = v + u$, where $v(x) = \varphi_2(x)/(2x)$ and

$$u(x) = \frac{1}{2}\log x + \frac{1-\gamma+\log 2\pi}{2} - xJ(x), \qquad J(x) = \int_x^\infty \varphi_2(t)\frac{\mathrm{d}t}{t^3}.$$

Throughout the proof, implied constants depend only on λ, and x and x' lie in $I = [\lambda/2, 3\lambda/2]$. Since φ_2 is continuous, J is differentiable with $J'(x) = -\varphi_2(x)/x^3$, and so

$$u'(x) = \frac{1}{2x} - J(x) + \frac{\varphi_2(x)}{x^2}.$$

The functions $1/(2x)$, J and x^{-2} are Lipschitz on I, and φ_2 is bounded. Hence $|u'(x) - u'(x')| \ll |x - x'| + \omega(|x-x'|;\varphi_2)$. It follows that

$$|\Delta^2 u(\lambda;h)| = \left|\int_0^h \big(u'(\lambda+s) - u'(\lambda-s)\big)\,\mathrm{d}s\right| \ll h\big(h + \omega(2h;\varphi_2)\big).$$

Next, $1/(2(\lambda\pm h)) = 1/(2\lambda) \mp h/(2\lambda(\lambda\pm h))$. Hence

$$\Delta^2 v(\lambda;h) = \frac{\Delta^2\varphi_2(\lambda;h)}{2\lambda} - \frac{h}{2\lambda}\left(\frac{\varphi_2(\lambda+h)}{\lambda+h} - \frac{\varphi_2(\lambda-h)}{\lambda-h}\right).$$

The expression in brackets is at most $|\varphi_2(\lambda+h) - \varphi_2(\lambda-h)|/(\lambda+h) + |\varphi_2(\lambda-h)|\cdot 2h/(\lambda^2-h^2)$ in absolute value, which is $\ll \omega(2h;\varphi_2) + h$. Combining the two estimates with (4) for $\delta = 2h \le 1/2$, we obtain the stated bound. □

Lemma 3 shows that a second difference of A is, up to an error of order $h^2\log(1/h)$, a second difference of φ_2 divided by 2λ. At the scales h_k of Theorem 1, Lemma 2 gives a main term of order h_k/q_k, which is much larger than $h_k^2\log(1/h_k)$ because $h_k < q_k^{-2}$. It remains to find infinitely many convergents with $y > 2$.

Lemma 4. *Suppose that λ is irrational, and put $y_k = 1/(q_k^2|\lambda - p_k/q_k|)$. Then $y_k = x_{k+1} + q_{k-1}/q_k$ for every $k \geq 0$, and $y_k > 2$ for infinitely many k.*

Proof. We apply the recurrence (8) to the finite continued fraction whose partial quotients are $a_0, a_1, \ldots, a_k$ and x_{k+1}. Its value is λ, and we get

$$\lambda = \frac{p_k x_{k+1} + p_{k-1}}{q_k x_{k+1} + q_{k-1}}.$$

By the last identity in (8), we find that

$$\lambda - \frac{p_k}{q_k} = \frac{p_{k-1}q_k - p_k q_{k-1}}{q_k(q_k x_{k+1} + q_{k-1})} = \frac{(-1)^k}{q_k(q_k x_{k+1} + q_{k-1})},$$

and so $y_k = (q_k x_{k+1} + q_{k-1})/q_k = x_{k+1} + q_{k-1}/q_k$.

Suppose first that $a_{k+1} \geq 2$ for infinitely many k. Since λ is irrational, $x_{k+1} > a_{k+1}$, and so $y_k > 2$ for each such k. Otherwise there is $k_0 \geq 1$ such that $a_k = 1$ for every $k \geq k_0$. For $k > k_0$ we then have $x_{k+1} = [1; 1, 1, \ldots] = (1+\sqrt{5})/2$, and (8) gives

$$\frac{q_{k-1}}{q_k} = \frac{q_{k-1}}{q_{k-1} + q_{k-2}} \geq \frac{1}{2},$$

since $q_{k-2} \leq q_{k-1}$. Hence $y_k \geq (1+\sqrt{5})/2 + 1/2 > 2$ for every $k > k_0$. □

We are now ready to prove Theorem 1.

Proof of Theorem 1. Fix an irrational $\lambda > 0$. Let k run through the infinitely many indices given by Lemma 4, and write $p = p_k$, $q = q_k$, $h = h_k$ and $y = y_k$. Since $q_k \to \infty$ and $h_k \to 0$, we may assume that $q \geq 2$ and $h \leq \min(\lambda/2, 1/4)$. We apply Lemma 2 with $d = \lambda - p/q$, so that $\delta = h$. Since $y > 2$,

$$\Delta^2\varphi_2(\lambda; h) \geq \frac{h}{q}\Big(2\log 2 - \frac{35\pi^2}{324} - \frac{1}{2q}\Big) \geq \frac{h}{q}\Big(\frac{3}{10} - \frac{1}{2q}\Big),$$

where we used that $2\log 2 - 35\pi^2/324 = 0.3201\ldots$. Next, $h = 1/(q^2 y) < q^{-2} \leq 1/4$. The function $x \mapsto x\log(1/x)$ is increasing on $(0, 1/e)$, and so $h\log(1/h) \leq q^{-2}\log q^2 = 2q^{-2}\log q$. Combining these estimates with Lemma 3, we find that

$$\Delta^2 A(\lambda; h) \geq \frac{h}{q}\Big(\frac{3/10 - 1/(2q)}{2\lambda} - 2C_\lambda\frac{\log q}{q}\Big).$$

The right-hand side is positive when q is sufficiently large. Since $q_k \to \infty$, (1) holds for all but finitely many of the indices given by Lemma 4. □

Proof of Corollary 1. Báez-Duarte, Balazard, Landreau and Saias [1] proved that A has a strict local maximum at every positive rational (see also [3, Section 2]). In the working document, this can be read off from [1, Proposition 98], which states that for $0 < |t| \le 1/(2q)$,

$$A(p/q + t) - A(p/q) = \frac{|t| \log |t|}{2p} + D_{\pm}(p, q)t + O_{p,q}(t^2),$$

where $D_{\pm}(p, q)$ depends only on p, q and the sign of t. The term $|t| \log |t|/(2p)$ is negative and dominates the linear term as $t \to 0$. By Theorem 1, A has no local maximum at any positive irrational. □

5. Concluding remarks

A quantitative form. The proof of Theorem 1 gives more than (1). For every irrational $\lambda > 0$ there is a constant $c_\lambda > 0$ such that

$$A(\lambda + h_k) + A(\lambda - h_k) - 2A(\lambda) \ge c_\lambda \frac{h_k}{q_k}$$

for all sufficiently large k with $y_k > 2$. Since $q_k^2 h_k < 1$, we have $h_k/q_k > h_k^{3/2}$, and so the second difference is at least $c_\lambda h_k^{3/2}$ along these k. If λ has bounded partial quotients, then $h_k \asymp q_k^{-2}$, and the two lower bounds are of the same order.

Local minima. Let $r_1 < r_2$ be positive rationals. Since A is continuous and has strict local maxima at r_1 and r_2, its minimum on $[r_1, r_2]$ is smaller than $A(r_1)$ and $A(r_2)$, and it is therefore attained at an interior point. This point is not rational, since a strict local maximum is not a local minimum. Hence every interval in $(0, \infty)$ contains an irrational point at which A has a local minimum.

Problem 2. *Determine the set of irrational $\lambda > 0$ at which A has a local minimum.*

Relation to differentiability. The criterion of Balazard and Martin [4, Théorème 1] shows that A is differentiable at many irrationals, for instance at every irrational with bounded partial quotients. At such a point a local maximum would force $A'(\lambda) = 0$, so first-order information alone cannot decide Problem 1. This is why the argument above works with second differences, which do not involve $A'(\lambda)$, and why it needs the explicit remainder in (6) rather than the asymptotic formula (2). Balazard and Martin [5] have studied approximate functional equations involving the Gauss map in a general setting. The renormalization $t \mapsto 1/(q^2 t)$ in (6), taken at the convergents of λ, is closely related to the Gauss map, and it would be interesting to know whether their methods give the sign of the second differences of A at other scales.

Acknowledgements. AI assistance was used in developing and checking the argument and in preparing the exposition; the author has verified the proof.

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